

(notes) 3.3 Average Rate of Change

Global minimum/maximum: the ultimate highest or largest value achieved

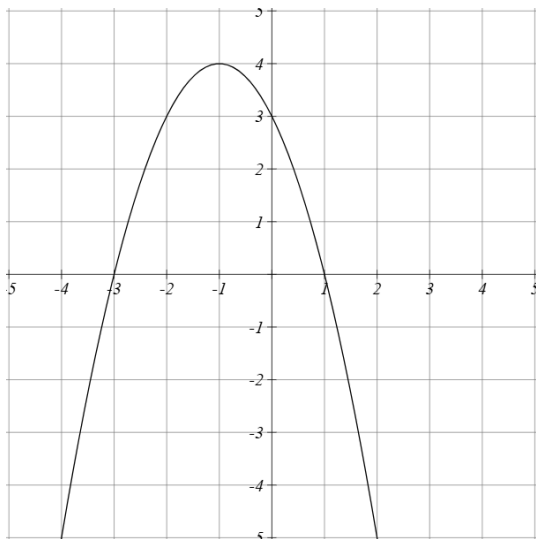
Local minimum/maximum: the lowest/highest point in a region

Hint with terms:

When we ask “**where**” on a function, we are asking for the x value or interval of x -values. Normally we ask where is the function zero (at certain x values), where is the function increasing or decreasing (an x interval)

When we ask for the “**value**” of a function, we are looking for the y -value. We can ask: what is the value of a function for some x -value? What is the minimum or maximum value of the function?

For the following graphs....



Where is the global max/min?

$x =$

What is the **value** of the global max/min?

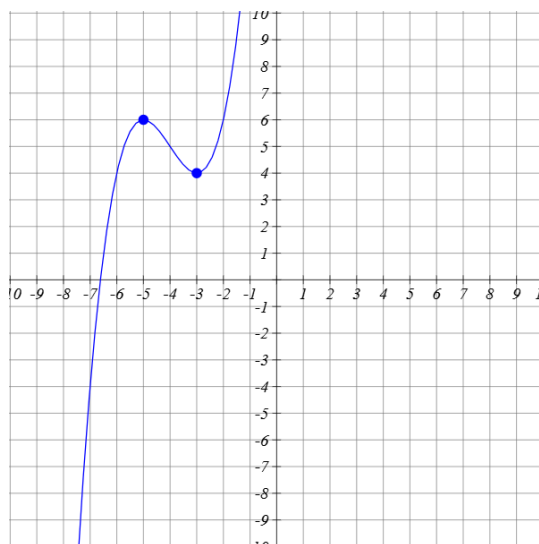
$y =$

Where is the graph **increasing**?

x -interval:

Where is the graph **decreasing**?

x -interval:



Does the graph have **global** or **local** max/min?

Where is the max? $x =$

Where is the min? $x =$

What is the **value** of the max? $y =$

What is the **value** of the min? $y =$

Where is the graph **increasing**?

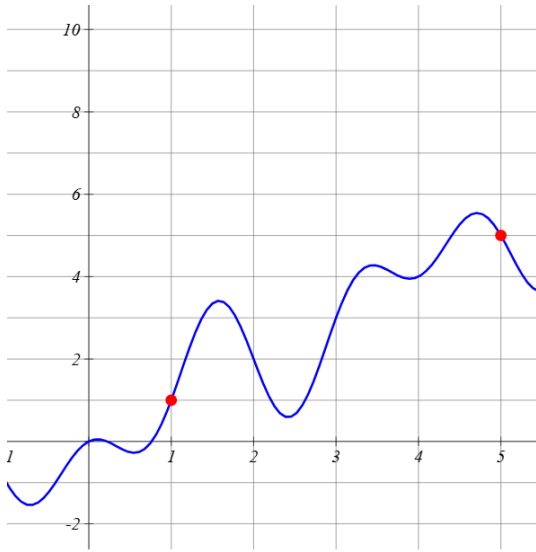
x -interval:

Where is the graph **decreasing**?

x -interval:

Average rate of change:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \rightarrow \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$



Find the average rate of change on the x -interval $[1, 5]$

Find the average rate of change on the function $f(x)$ on the x -interval $[-1, 2]$

$$f(x) = -4x^3 - 5$$

(Tricky) Find the average rate of change on the x -interval $[1, t]$

$$f(x) = 9x^2 - 4$$